

Total No. of Questions : 8]

SEAT No. :

**PB2381**

[Total No. of Pages : 2

**[6263] 231**

**B.E. (Mechanical Engineering)**

**PROCESS EQUIPMENT DESIGN**

**(2019 Pattern) (Semester - VIII) (Elective - VI) (402051A)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

**Q1) a)** Explain design of thick cylinders subjected to internal pressure based on maximum principal strain theory and derive equation to find thickness of cylinder wall. **[9]**

b) Explain the purpose of pre-stressing of cylinders. Explain autofrettage in detail with neat labeled sketch. **[9]**

OR

**Q2) a)** A long thick steel tube of inner diameter 150 mm is subjected to an internal pressure of 20 N/mm<sup>2</sup>. The allowable compressive and tensile stresses for the tube are 160 N/mm<sup>2</sup> and 100 N/mm<sup>2</sup> respectively. Determine the outer diameter of the tube using: **[10]**

- i) Maximum principal stress theory
- ii) Maximum shear stress theory

Derive the expression you use.

b) What are different types of supports used in horizontal and vertical pressure vessels? Explain bracket support with neat labeled sketch. **[8]**

**Q3) a)** Explain the three types of floating roofs used in floating roof type tank with sketches. **[9]**

b) Explain the pump selection procedure for pump used in process industry. **[8]**

OR

**P.T.O.**

- Q4)** a) Explain shell and tube type of heat exchanger with neat labeled sketch. [8]  
b) List the different types of valves used on pipe line. Explain any two types of valves with neat sketch. [9]

- Q5)** a) List the different pressure measuring instruments and explain principal, construction and working of Bourdon tube with neat labeled sketch. [9]  
b) Discuss with neat labeled sketch, any type of flow meter used to measure flow of highly corrosive liquids involving measurement of erosive slurries. [9]

OR

- Q6)** a) Explain function of controller. [4]  
b) State variables of Automatic Process control. [6]  
c) Describe with neat sketches, Principal, construction and working of LVDT. [8]
- Q7)** a) What is purpose of using protective coating on metal? Write note on Metallic coating and inorganic coating. [8]  
b) List NDT methods used for surface defects. Discuss any two methods used for detection of surface defects. [9]

OR

- Q8)** a) Explain how AI/IOT can be used to overcome challenges at Fuel Pumping station. [8]  
b) Differentiate between Destructive and Non-destructive Testing. Explain Radiographic testing used for detection of internal defects. [9]

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